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pyRTX: a Python package high precision computation of non gravitational forces on deep space probes

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Summary

With the constant improvement of radiometric tracking systems, inaccuracies in the non-gravitational force modeling have become one of the limiting factors to precise orbit determination, and the scientific products that it enables. The main factor impacting the limited accuracy of non-gravitational force models is the complex 3D shape of the spacecraft. While fast, reliable, analytical models are available for simple shapes (spheres, cubes, etc), no such model is generally available for a complex shape. This software package aims to address this limitation by legeraging ray-tracing to compute the complex interaction between the forcing environment (radiation, atmosphere) and the three dimensional shape of the spacecraft.

Statement of need

Several scientific investigations require high-precision reconstruction of spacecraft trajectories. Among these, one of the most demanding is the determination of the gravity field of Solar System bodies (planets, moons). This task is accomplished by solving the so-called orbit determination (OD) problem (Milani & Gronchi, 2009; Tapley et al., 2004). The solution of the OD in the adjustment of a dynamical model (a set of differential equations) describing the spacecraft motion. Systematic errors in the dynamical model will almost inevitably lead to systematic errors in the solution. In the recent years significant improvements in radiometric tracking system, have led to more and more precise measurements of the spacecraft position and velocity (the input to the OD), thus requiring increasingly more accurate dynamical models (Asmar et al., 2005; Cappuccio et al., 2020; Mazarico et al., 2023). One of the major limitations of current dynamical modelling of deep-space probes consists in the complex interaction between the spacecraft shape and the atmosphere, and with radiative forces (solar radiation pressure, albedo, thermal infrared radiation). We developed the pyRTX software package to address this limitation. Leveraging the ray-tracing technique, originally developed in computer graphics, instead of relying on simplified macro-models (flat plates, cylinders, etc) pyRTX is able to use the actual 3D shape of the spacecraft (provided as, for example, an .obj file).

Mathematics

Single dollars ($) are required for inline mathematics e.g. $f(x) = e^{\pi/x}$

Double dollars make self-standing equations:

$$\Theta(x) = \left\{\begin{array}{l} 0\textrm{ if } x < 0\cr 1\textrm{ else} \end{array}\right.$$

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You can also use plain LaTeXfor equations

$$\hat f(\omega) = \int_{-\infty}^{\infty} f(x) e^{i\omega x} dx \tag{1}$$

and refer to Equation 1 from text.

Citations

Citations to entries in paper.bib should be in rMarkdown format.

If you want to cite a software repository URL (e.g. something on GitHub without a preferred citation) then you can do it with the example BibTeX entry below for Smith et al. (2020).

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Figures

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